

84.1.

P1. Solution:
$$\begin{cases} x - x^2 - 2xy = 0 \\ 2y - 2y^2 - 3xy = 0 \end{cases}$$

$$\Rightarrow x(1-x-2y) = 0 \Rightarrow x=0 \text{ or } 1-x-2y=0$$

$$y(2-2y-3x) = 0$$

i) When $x=0$, we have

$$y(2-2y) = 0 \Rightarrow y=0 \text{ or } y=1$$

ii) When $1-x-2y=0$, we have.

$$y(2-2y-3(1-2y)) = 0$$

$$y(-1+4y) = 0 \Rightarrow y=0 \text{ or } y=\frac{1}{4}$$

$$\downarrow \quad \downarrow$$

$$x=1 \quad x=\frac{1}{2}$$

Thus, this differential equation has 4 equilibrium values.

$$(0,0), (0,1), (1,0), (\frac{1}{2}, \frac{1}{4})$$

$$\beta \neq 0, \mu \neq 0$$

P2. Solution:
$$\begin{cases} -\beta xy + \mu = 0 \\ \beta xy - \gamma y = 0 \Rightarrow y(\beta x - \gamma) = 0 \\ \Rightarrow y=0 \text{ or } x = \frac{\gamma}{\beta} \end{cases}$$

i) When $y=0$, we have

$$-\beta xy + \mu = 0 + \mu \neq 0$$

ii) When $x = \frac{\gamma}{\beta}$, we have.

$$-\beta \cdot \frac{\gamma}{\beta} \cdot y + \mu = 0 \Rightarrow y = \frac{\mu}{\gamma}$$

Thus, this equation has one equilibrium value $(\frac{\gamma}{\beta}, \frac{\mu}{\gamma})$

P3. Solution:
$$\begin{cases} ax - bxy = 0 \Rightarrow x(a - by) = 0 \\ -cy + dx = 0 \Rightarrow y(-c + dx) = 0 \\ z + x^2 + y^2 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x=0 \text{ or } y = \frac{a}{b} \\ y=0 \text{ or } x = \frac{c}{d} \\ z = -x^2 - y^2 \end{cases}$$

$$a \neq 0, b \neq 0, c \neq 0, d \neq 0$$

Thus, this equation has $\geq$ equilibrium values

$$(0,0,0), (\frac{c}{d}, \frac{a}{b}, \frac{c^2}{d^2} - \frac{a^2}{b^2})$$

P5. Solution:
$$\begin{cases} xy^2 - x = 0 \Rightarrow x(y^2 - 1) = 0 \\ x \sin \pi y = 0 \Rightarrow x=0 \text{ or } \sin \pi y = 0 \end{cases}$$

$$\Rightarrow x=0 \text{ or } y = \pm 1$$

i) When $x=0$, for any $y=y_0$, we have.

$$x(y^2 - 1) = 0, x \sin \pi y = 0$$

ii) When $y = \pm 1$, $\sin(\pm \pi) = 0$, so for

any $x=x_0$, we have.

$$x(y^2 - 1) = 0, x \sin \pi y = 0$$

Thus, this equation has the following equilibrium values.

$$(0, y_0), (x_0, 1), (x_0, -1)$$

where x_0, y_0 are arbitrary.

P6. Solution:
$$\begin{cases} \cos y = 0 \Rightarrow y = k\pi + \frac{\pi}{2}, k=0, \pm 1, \dots \\ \sin x - 1 = 0 \Rightarrow x = 2k\pi + \frac{\pi}{2}, k=0, \pm 1, \dots \end{cases}$$

Thus, this equation has the following equilibrium values

$$(2k\pi + \frac{\pi}{2}, k\pi + \frac{\pi}{2}), k=0, \pm 1, \dots$$

P9. Solution: i)
$$\begin{cases} ax + by = 0 \\ cx + dy = 0 \end{cases}$$

$$ad - bc \neq 0 \Leftrightarrow \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq 0 \Leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ is nonsingular}$$

$$\Leftrightarrow x=y=0$$

ii) If $ad - bc = 0$, then $(ax + by) = \tilde{c}(cx + dy)$ for some constant $\tilde{c}$. Hence.

$$ax + by = 0 \Rightarrow y = -\frac{ax}{b}$$

Then, $(x_0, -\frac{ax_0}{b})$ is an equilibrium value for arbitrary x_0 and these points form a line.